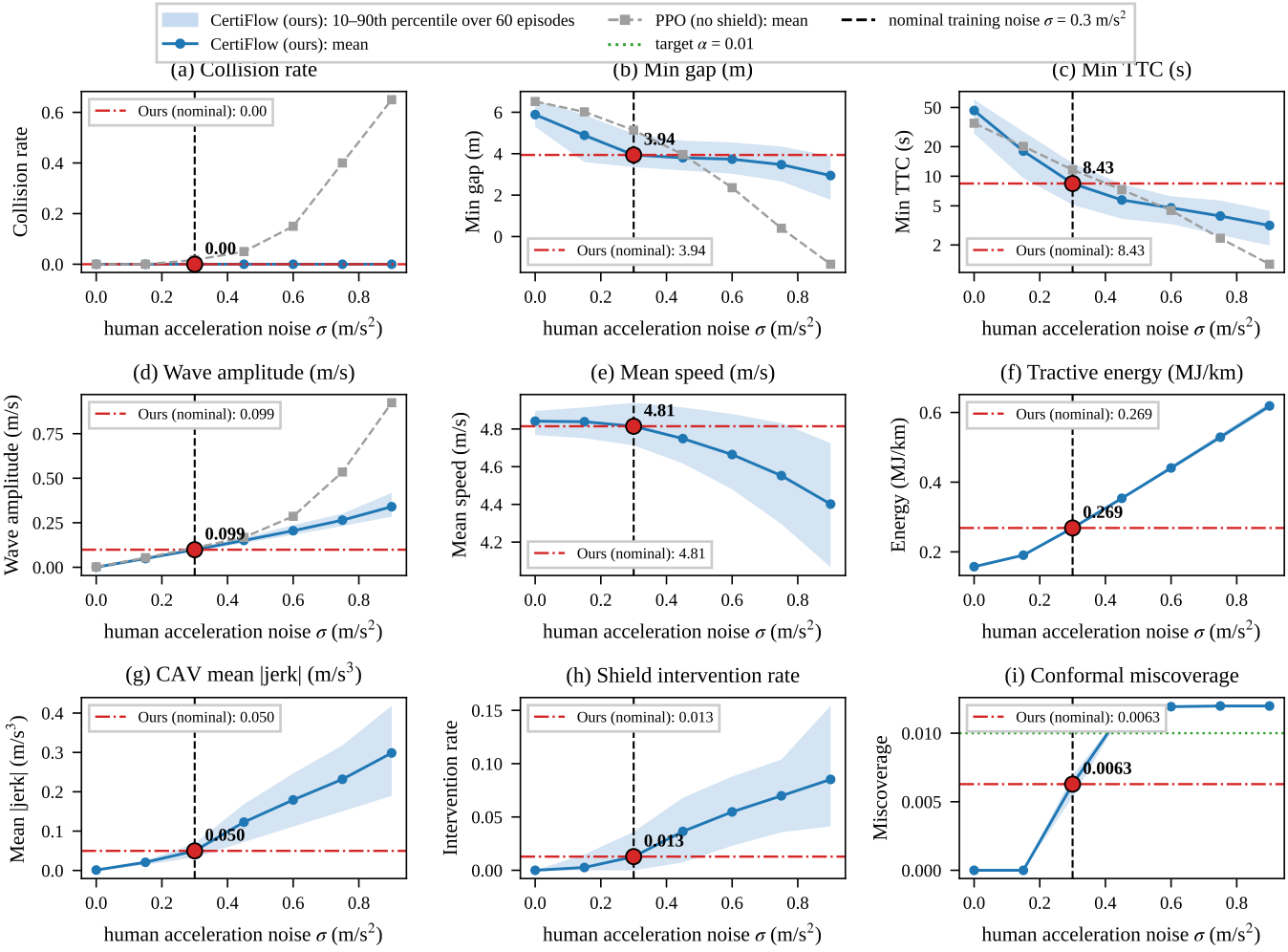
CertiFlow (ours): 10–90th percentile over 60 episodes
CertiFlow (ours): mean
PPO (no shield): mean
target α = 0.01
nominal training noise σ = 0.3 m/s^2
(a) Collision rate
Collision rate
Ours (nominal): 0.00
0.00
(b) Min gap (m)
Min gap (m)
3.94
Ours (nominal): 3.94
(c) Min TTC (s)
Min TTC (s)
8.43
Ours (nominal): 8.43
(d) Wave amplitude (m/s)
Wave amplitude (m/s)
Ours (nominal): 0.099
0.099
(e) Mean speed (m/s)
Mean speed (m/s)
4.81
Ours (nominal): 4.81
(f) Tractive energy (MJ/km)
Energy (MJ/km)
Ours (nominal): 0.269
0.269
(g) CAV mean |jerk| (m/s^3)
Mean |jerk| (m/s^3)
Ours (nominal): 0.050
0.050
(h) Shield intervention rate
Intervention rate
Ours (nominal): 0.013
0.013
(i) Conformal miscoverage
Miscoverage
Ours (nominal): 0.0063
0.0063
human acceleration noise σ (m/s^2)